

An introduction to TIPS and US CPI derivatives

2026 update

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- The TIPS market remains the largest inflation-linked bond market in the world, with \$2.2tn outstanding—larger than the UK and Euro area linker markets combined
- We provide an overview of the TIPS market and the mechanics of inflation-adjusted cash flows, including considerations around CPI seasonality, the embedded floor option, the drivers of inflation breakevens, and the sources of demand in the market. We also discuss inflation derivatives, including zero-coupon swaps, asset swaps, and options
- Gross issuance is likely to total \$235bn in 2026, over seven times its 1997 level. While TIPS have fallen as a share of the Treasury market, Treasury remains committed to the TIPS program and is likely to maintain TIPS allocations as a moderate proportion of the debt stock
- TIPS breakevens reflect not only inflation expectations, but also inflation risk premium and liquidity premium components. We model the fair value of seasonally- and carry-adjusted breakevens based on commodity prices, implied volatility (VIX), economists' inflation expectations and the expected forward path of Fed policy
- Private rather than official investors have become the dominant holders of TIPS. We expect the Fed's share of the TIPS market to continue to decline as the Fed shifts the duration of its portfolio lower by increasing its bill share. Foreign private and official institutional shares of the TIPS market have converged to roughly 19%, which represents a record high for the private community but represents a material decline for the official community from a peak of 32% in 2015
- Liquidity in the US CPI derivative market is growing, but is still not as widely traded as cash linkers; daily trading volumes in USD inflation swaps have grown from \$3bn in 2021 to roughly \$11.6bn in 1H26, representing 46% of TIPS daily trading volumes. We provide an overview of the most traded CPI derivatives and how to screen for relative value

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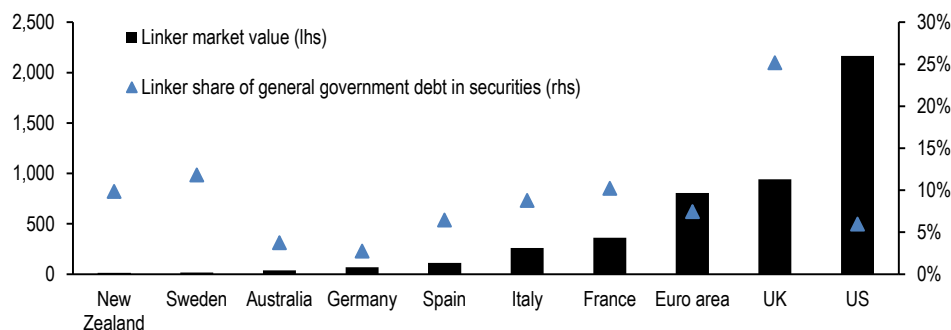
See page 26 for analyst certification and important disclosures.

Introduction

Treasury began issuing Treasury Inflation-Protected Securities (TIPS) in January 1997, and TIPS have grown to be the largest inflation-linked bond market in the world, with \$2.2tn outstanding as of June 2026, including \$0.38tn of inflation compensation (**Figure 1**). Over the following pages, we provide an overview of the TIPS market, the mechanics of inflation-adjusted cash flows, including considerations around CPI seasonality and the embedded floor option, the drivers of inflation breakevens, and the sources of demand in the market. We also discuss inflation derivatives, including zero-coupon swaps, asset swaps, and options.

Figure 1: The TIPS market is the largest linker market in the world

Amount of linker debt outstanding* (lhs, USD Bn) relative to linker share of general government debt in securities** (%)



* The uplifted notional outstanding for a given linker is the product of its notional outstanding by its indexation factor. Note: France include FCPIxT- and HICPxT-linked bonds, Italy excludes BTP Italia, and Euro area is the aggregate of France (FCPIxT and HICPxT), Germany (HICPxT), Italy (HICPxT) and Spain (HICPxT).

**General government debt in securities used for a consistent source across markets. Euro area references the 4 linker markets (France, Italy, Spain and Germany).

For more information on global inflation markets, see [Inflation markets - concepts and themes](#), Francis Diamond, 04/29/26)

Source: US Treasury, national DMOs, IMF, Haver Analytics

Under the current issuance calendar, Treasury auctions two new 10-year TIPS per year (January and July maturities) and two new 5-year issues (April and October maturities), and each is reopened twice. Treasury auctions one new 30-year issue per year (February maturity) followed by one reopening auction (**Figure 2**).

Gross issuance is likely to total \$235bn in 2026, over seven times its 1997 level, and roughly in-line with 2025 (**Figure 3**). Alongside an expectation that nominal coupon auction sizes hold steady, we forecast that Treasury would maintain current TIPS auction sizes over the coming quarters and would revisit the optimal TIPS share at a future refunding. TIPS share of outstanding debt is likely to decline further to 6.6% by the end of the year, below the bottom of the optimal 7-9% range that has been recommended by the TBAC in the past. The [debt optimization model](#) presented by the Treasury Borrowing Advisory Committee (TBAC) in 2022 showed that debt service costs for TIPS are generally lower than similar maturity nominal issuance. However, the volatility of the deficit is higher when issuing TIPS relative to nominals, particularly for 30-year TIPS. In 2018, TBAC found that TIPS offered diversification benefits for Treasury due to the negative correlation between CPI and US primary deficits. An extension of that model presented in August 2020 showed that while FRNs and bills can also reduce risk due to the negative correlation between short rates and primary deficits, TIPS may provide a greater reduction in rollover risk at the zero lower bound (ZLB). Thus, Treasury is likely to remain committed to the TIPS program and maintain TIPS allocations as a moderate proportion of the debt stock.

Figure 2: Treasury conducts monthly auctions of TIPS

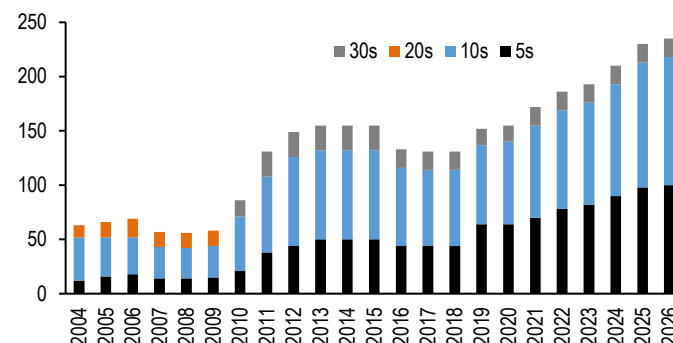
J.P. Morgan forecast for gross issuance of TIPS in 2026; reopening shaded in grey; \$bn of real principal

	5y TIPS	10y TIPS	30y TIPS	Subtotal
Jan 26		21		
Feb 26			9	
Mar 26		19		49
Apr 26	26			
May 26		19		
Jun 26	24			69
Jul 26		21		
Aug 26			8	
Sep 26		19		48
Oct 26	26			
Nov 26		19		
Dec 26	24			69
2026 Total	100	118	17	235

Source: US Treasury, J.P. Morgan

Figure 3: Gross issuance of TIPS has increased seven-fold from its 1997 levels

Historical gross issuance of TIPS by maturity and J.P. Morgan forecast for 2026; \$bn of real principal

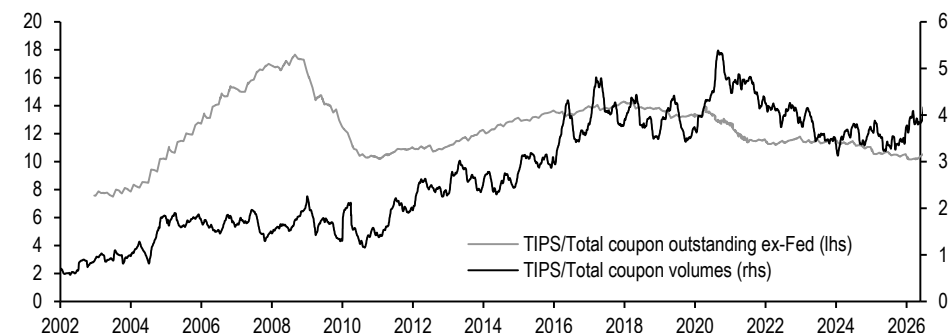


Source: US Treasury, J.P. Morgan

Although the TIPS market has been around for almost thirty years and comprises a significant share of the Treasury market, TIPS remain far less liquid than their nominal counterparts. Over recent years, daily trading volumes with primary dealers have risen to an average of \$24bn/day in 2026, but equate to about 1% turnover each day. In contrast, daily trading volumes in nominal coupon Treasuries have been around \$606bn/day, or about 3% turnover, and TIPS have represented roughly 4% of total coupon volumes (**Figure 4**). The liquidity in TIPS relative to nominals has decreased since 2021 as TIPS issuance has not increased as quickly as it did for nominal notes and bonds. Additionally, a greater share of buy-and-hold investors likely contributes to the lower turnover (see “Who trades TIPS” below).

Figure 4: TIPS are not as liquid as nominal Treasuries

Average daily TIPS trading volumes as a fraction of total coupon Treasury volumes (LHS) and TIPS outstanding (ex-Fed) as a fraction of total coupon Treasuries outstanding (ex-Fed) (RHS); % both axes



Source: US Treasury, Federal Reserve Bank of New York

TIPS are typically auctioned on the third Thursday of the month at 1:00 p.m., with the announcement of the auction occurring on the previous Thursday at 11:00 a.m. The auction process works the same as all Treasury auctions: an investor may either submit a non-competitive or competitive bid for a TIPS auction, where competitive bids are subject to a Dutch auction process (see [The Hitchhikers Guide to the Treasury Market](#) for more detail), and the coupon rate is floored at 0.125%. When real yields on TIPS are

trading negative, as they were in 2020 and 2021, this rule means that new-issue TIPS are auctioned at a premium (i.e., the initial price is greater than 100).

Overview of TIPS

TIPS are structured to provide inflation-protected cash flows to investors. Thus, an investor that purchases a TIPS outright locks in a real return that is not a function of future inflation. Inflation protection is achieved by indexing the principal of a TIPS to the US headline CPI non-seasonally adjusted series (CPURNSA Index on Bloomberg) published by the Bureau of Labor Statistics (BLS)—the principal of a TIPS increases with inflation and decreases with deflation. Each day, the inflation-adjusted principal is equal to par multiplied by the inflation index ratio, where the index ratio is calculated as the reference CPI on the trade settlement date divided by the reference CPI on the first issue date (the base CPI value).

Since CPI for a given month is released about two weeks after the end of that month, and CPI must be known prior to issuing a TIPS, the **principals are indexed to CPI with a 3-month lag**. Specifically, the reference CPI on the first day of a calendar month is based on the CPI for the third preceding calendar month. For example, the principal on June 1 is based on the CPI reported for March, and the principal on July 1 is based on April CPI. All rounding is to 5 decimal places. On any day between June 1 and July 1, the reference CPI value is calculated by linearly interpolating between the March and April CPI values.

Semiannual coupon payments are calculated as a fixed percentage of the inflation-adjusted principal. **At maturity, holders of TIPS are paid the adjusted principal or the original principal (par), whichever is greater.** Effectively, this means that TIPS have an embedded floor on the principal amount.

TIPS are quoted on a **real price/real yield basis**, and real yields are derived in the same way as fixed-coupon nominal Treasuries, using the “real” cash flows (real price, real coupons, and par at maturity) to calculate a real Internal Rate of Return (“IRR”). When making a cash transaction in TIPS, the real price of the bond (plus the real coupon accrued) is adjusted by the daily index ratio.

A list of daily TIPS index ratios can be found on [Treasury's website](#) and is available in our daily [TIPS Fitted Curve Relative Value Report](#). One can also find the index ratio for a particular bond using the Bloomberg YA screen.

Example: The current index ratio for the 1.25% Apr-31 TIPS, issued on April 30, 2026, for a transaction settling on June 15, 2026, is calculated as follows:

Reference CPI for June 15, 2026 = Interpolated value between Ref March 2026 (330.213) and Ref April 2026 (333.020) = $330.213 + 15/30 * (333.020 - 330.213)$
= 331.6165

Base CPI value = reference CPI for April 30, 2026 = 325.9674

Index Ratio = reference CPI for June 15, 2026/ reference CPI for April 30, 2026
= $331.6165 / 325.9674$
= 1.01733

In this case, because headline inflation was positive through 2026, the principal on the Apr-31 TIPS in June 2026 was higher than it was at issue.

Carry is defined as the income accrued while holding a security minus the cost of funding the position. In other words, it measures the total return over a particular timeframe assuming that the IRR of the investment is unchanged (i.e. no capital gains or losses from changes in the IRR). In the case of conventional fixed-coupon bonds, this can be seen in simple terms as the coupon accrued minus the cost of funding the bond investment (i.e. its repo rate). Alternatively, one can express carry as the spot price of the bond minus its forward price.

We can also analyze carry for inflation-linked bonds in this manner. **The difference is that the dirty price of a linker includes the accrual of a real coupon plus the accrual of inflation for the given horizon.** Since the cash flows of inflation-linked bonds are indexed, this means that there is a positive carry effect during periods when (lagged) inflation indices increase and a negative carry effect when indices decrease. The actual carry is thus the real coupon accrued plus inflation accrual minus the cost of funding. Effectively, this means that forward prices of linkers depend on real yields, funding rates, and the evolution of inflation indices. The normal evolution of inflation indices is what makes carry in inflation linked bonds particularly interesting. The seasonal variation in inflation, discussed in more detail later in this note, makes carry on inflation-linked bonds very volatile, switching from positive to negative throughout the year. An example of a detailed carry calculation is provided in the Appendix.

Note that when calculating forwards we must know the value of the indexation factor at the horizon of choice, or use forecasts. Given the lag used in the inflation accrual calculation (3-month lag for most bonds) and the publication schedule (most inflation prints come out around the middle of the following month) it is always possible to know the actual value of the indices for the current month. Once a CPI print is released, one can calculate the forwards for the next calendar month. For example, the May 2026 CPI report was released on June 10, 2026. Up until that point, we only knew the CPI reference values through July 1, 2026 (based on the April 2026 CPI report). In other words, we could only calculate the carry for about 2 weeks with certainty. Once the May CPI report was released, however, we knew the CPI reference values through August 1, 2026, which meant we could calculate carry for over 1 month with certainty.

The inflation breakeven

Since TIPS provide inflation-protected cashflows, we can calculate breakeven inflation rates across the term structure by comparing TIPS real yields with nominal Treasury yields. Breakeven inflation can simply be thought of as the average annual inflation needed over the life of the TIPS in order for it to provide the same return as a Treasury with the same maturity (i.e., the level of inflation that makes an investor indifferent between buying TIPS or Treasuries). We approximate this breakeven rate as the similar-maturity Treasury yield minus TIPS real yield (usually quoted in basis points).

The Fisher Equation expresses the relationship between nominal rates, real rates and inflation by mathematically decomposing nominal rates into real rates and inflation. The Fisher Equation is expressed as follows:

$$(1 + n) = (1 + i) \times (1 + r)$$

where: n = nominal rates

i = inflation rate

r = real rate

If we expand this out we can see that:

$$1 + n = 1 + i + r + ir$$

As ir is generally a very small term in this equation, the Fisher equation is often shown in its simplified approximate form:

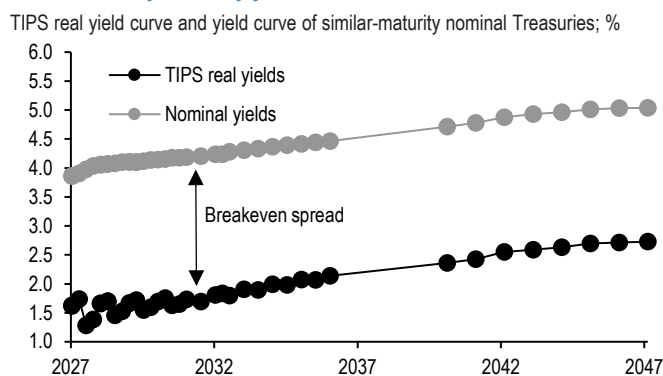
$$n = i + r$$

In fact, in the bond sphere, breakeven inflation is calculated simply by subtracting the real yield from the nominal yield of a comparator bond.

Figure 5 illustrates the breakeven curve. Note that because breakeven rates are annualized rates, the magnitude of front-end (Y to maturity) breakevens can be quite large. For example, a breakeven of 490bp for a TIPS with two months to maturity implies that the market is pricing 0.4% monthly gain in inflation over the next two months (on a lagged basis). Similarly, a breakeven of -350bp for a TIPS with two months to maturity implies the market is pricing in -0.3% monthly declines in inflation over the next two months (on a lagged basis): $-3.5\% = (1 - 0.3\%)^{12} - 1$.

Nominal and real yields can be decomposed into various factors as shown in **Figure 6**. Although inflation breakevens are a useful measure of the market expectations of inflation, other market-driven factors also influence breakeven rates. Specifically, the inflation breakeven can be rewritten as expected inflation plus an inflation risk premium, and each factor can vary across maturity points. Real TIPS yields reflect a liquidity premium (or illiquidity cost) to compensate investors for the relative illiquidity of inflation-linked bonds compared to nominals. As we saw during the 2008 financial crisis and again in March 2020, this component can sometimes become a dominant driver of breakevens (see “Drivers of breakevens” below).

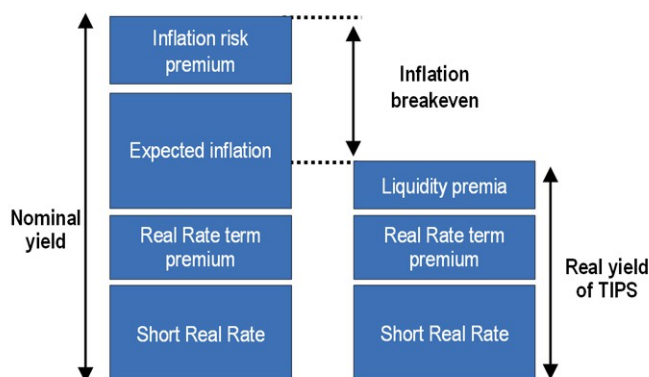
Figure 5: The breakeven is the spread between a TIPS real yield and a similar-maturity Treasury yield...



Source: J.P.Morgan

Figure 6: ...and this “breakeven” rate reflects more than just expected inflation

Breakdown of nominal and real TIPS yields



Source: J.P.Morgan

Still, market participants, as well as the Federal Reserve, use TIPS breakevens as a market-based measure of inflation expectations. In particular, investors and the Federal Reserve regularly monitor the 5Yx5Y forward breakeven measure derived from par Treasury rates. Compared to spot breakevens, the Fed views 5Yx5Y breakevens as less affected by cyclical factors such as energy prices and thus a better measure of how well the market thinks it is meeting its goal of longer-term price stability (see [Dropping anchor: Why five-year forward five-year inflation has stayed rangebound](#), 06/09/26 and [Why we look at 5-year forward 5-year inflation](#), 3/23/11).

A breakeven position is established by buying (selling) TIPS and selling (buying) a DV01-matched amount of Treasuries. Usually a 1:1 risk ratio is used, but other ratios can be used if the investor believes TIPS will be more/less volatile than Treasuries. For example, investors who want to position for the outperformance of TIPS while limiting their duration exposure may choose to weight the short nominal Treasury leg of a breakeven widener trade, based on the empirical beta of real yields versus nominal yields. In addition to trading breakevens themselves, investors can also take views on the breakeven curve, or do breakeven switches.

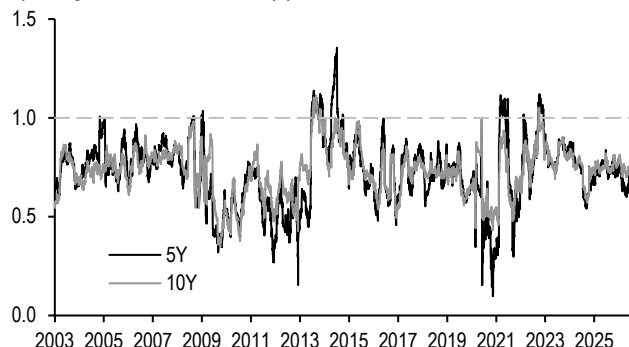
Drivers of TIPS and breakevens

Government bond markets and nominal yields: The dominant driver of real yields is nominal yields, with real yields tending to increase when nominal yields increase and vice-versa. Historically, the beta between TIPS yields and nominal yields has generally been below 1, meaning that TIPS yields rise less than nominal yields in a sell-off, and decline less in a rally (**Figure 7**). As a result, breakevens typically widen in a sell-off and narrow in a rally. This makes intuitive sense because stronger growth expectations generally correlate with stronger inflation expectations and higher nominal yields, while weaker growth and inflation expectations tend to push nominal yields lower.

While this directionality has been consistently observed over the life of the TIPS product, there have been periods where this relationship has broken down. The beta between real yields and nominal yields, especially in the 5-year sector, declined sharply in the 2011-2012 period when enhanced forward guidance was last introduced at the effective lower bound: 5-year nominal yields were little changed between August 2011 and the end of 2012, while real yields continued to drift lower. During the “taper tantrum” of 2013, real/nominal yield betas rose above 1, and breakevens narrowed during the sell-off, likely due to a sharp rise in the liquidity premium as market depth declined. Additionally, in 2020, with the Fed back at the lower bound, nominal yield betas dropped across the curve, to the lowest levels on record, and picked up above 1 as the world gradually re-opened post the announcement of a COVID vaccine in November 2020. The beta again moved above one in 2022, as inflation no longer seemed transitory and the Fed embarked on a 525bp hiking cycle over the following 15 months.

Figure 7: : The beta between TIPS yields and nominal yields is generally less than 1 and is now back in line with averages

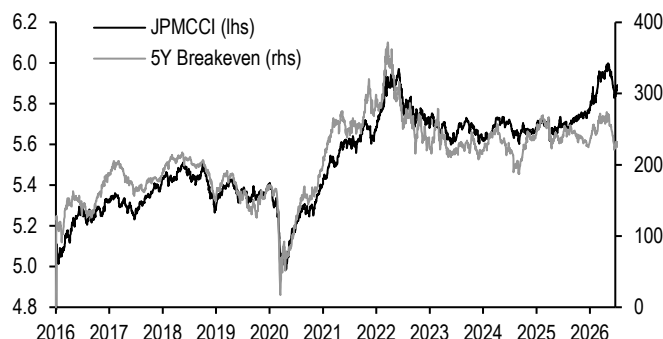
Rolling 3-month betas of daily changes in on-the-run TIPS yields regressed against daily changes in on-the-run Treasury yields;



Source: J.P. Morgan

Figure 8: Front-end inflation expectations tend to be largely driven by commodity prices

5-year US CPI inflation swap rate (lhs; %) versus log of J.P.Morgan Commodity Curve Index price* (rhs)



* We show the logarithm of the commodity price so that the scale reflects percentage changes rather than index level changes, since the magnitude of the percentage change influences the inflation rate

Source: J.P.Morgan

Inflation prints and expectations: The evolution of actual headline and core CPI inflation prints generally heavily influence expectations of future inflation. In addition, factors affecting expectations of future inflation are also important drivers of TIPS and breakevens. Near-term inflation dynamics also determine the carry offered by TIPS, given their indexation mechanics. Commodity prices tend to drive headline inflation and thus heavily influence front-end TIPS (**Figure 8**). When the month-on-month change in the CPI is large, so is the carry offered by TIPS, and vice versa. It is common for investors to hedge the energy exposure on a front-end TIPS or breakeven trade by selling oil or gas futures against it. We discuss energy hedging breakeven exposure in more detail in [De-energizing inflation breakevens](#), 4/17/20. When we model inflation we simplify those factors into 4 variables:

- **1) JPMorgan Commodity Curve Index (JPMCCI)** - Since 2022, we have included a broad commodity curve in our model given that TIPS breakevens have displayed a higher sensitivity to broad commodity prices than to energy prices alone (see [US Fixed Income Markets 2023 Outlook](#), 11/22/2022). The J.P. Morgan Commodity Curve Index seeks to track the performance of futures across the most investable parts of the commodity curve. The aggregate index is comprised of 33 commodity indices, grouped into five sectors - Energy, Industrial Metals, Precious Metals, Agriculture, and Livestock.
- **2) JPMCCI squared** - Captures the non-linear sensitivity of energy prices to inflation, in part because large increases in energy prices have proven temporary, diminishing the signal from the price increase.
- **3) OIS Curve (3Mx3M/15Mx3M OIS curve)** - The forward expectations for the path of Fed policy. The OIS curve shape has been a good proxy for Fed tightening and easing cycles. Fed policy is closely linked to core inflation rather than the headline inflation which is primarily influenced by commodity prices.
- **4) Economists' minus CPI implied inflation expectations** - We partially account for inflation expectations through the market pricing of the OIS curve, however, this does not account for economists' expectations. We include an additional variable that looks at the instances where market and economists' expectations may differ. We previously developed a model for forecasting inflation expectations based upon

realized inflation (see [A Fair Value for US Bonds, Credit and Equities](#), Nikolaos Panigirtzoglou, 06/30/2005). Anchoring is the lack of sensitivity of long-term inflation expectations to fluctuations in near-term inflation. To define a variable to account for economists' expectations, we use the difference between current expectations for CPI over the next 10 years from the SPF minus what our model would estimate that to be based on current 2- and 10-year CPI.

Liquidity risk (VIX Index): Measure of implied volatility that captures changes in the demand for liquidity. Technically, TIPS have the same sovereign credit risk as nominal Treasuries, since they are both obligations of the US federal government. However, in practice, TIPS often trade like risky assets, with breakevens widening (TIPS outperforming) on “risk-on” days, and breakevens narrowing (TIPS underperforming) on “risk-off” days. One explanation is that TIPS—like several other risky asset sectors—are significantly less liquid than Treasuries. Often, flight-to-quality events are also flight-to-liquidity events, so if investor risk appetite declines—even if inflation expectations are unchanged—TIPS may underperform and breakevens may narrow.

Putting these factors together, we present our fair value model for inflation markets in **Figure 9**: We regress 1-month forward, seasonally-adjusted breakevens and forward zero-coupon inflation swaps on the 5 variables above. (We discuss the seasonal adjustment process in the following section.) Our model implies that breakevens tend to widen as commodity prices rise but with diminishing returns, such that beyond a certain point further commodity price increases can lower breakevens. Additionally, the model implies that as volatility spikes, breakevens narrow given the lower liquidity of nominal Treasuries. Furthermore, the positive beta to the slope of future Fed pricing implies that both breakevens and the Fed react to incoming firm or soft economic data. Finally, as realized inflation rises/falls relative to economists' inflation expectations, anchoring helps to offset that impact, hence the negative beta between the variable and inflation breakevens. We use fair value models to discern richness/cheapness and help inform our outlook for breakevens.

Figure 9: Our inflation fair value model shows breakevens are largely driven by commodity prices, Fed policy expectations, risk appetite, and economist's inflation expectations

Statistics for our breakeven and zero-coupon inflation swap fair-value model* (bp), regression over the past 7 years using daily data

	JPMCCI		JPMCCI ²		3m3m/15m3m OIS curve, bp		VIX Index		Economists' inflation expectations		Intercept (bp)	R-sq	Std error (bp)	Fair value	Residual	Current
	Beta	T-stat	Beta	T-stat	Beta	T-stat	Beta	T-stat	Beta	T-stat						
5Y	4.01	60.86	-0.0060	-52.17	0.32	35.09	-0.43	-7.08	-0.05	-3.47	-391	91%	14.6	268	-43	225
10Y	2.84	62.02	-0.0043	-53.44	0.19	30.14	-0.35	-8.24	-0.11	-10.68	-220	92%	10.1	248	-25	223
30Y	2.06	49.09	-0.0032	-43.24	0.10	16.69	-0.21	-5.47	-0.12	-12.45	-99	88%	9.3	235	-17	218
5y5y	1.50	37.01	-0.0024	-33.22	0.08	13.52	-0.28	-7.46	-0.23	-24.05	9	86%	9.0	254	-16	238

*Fair value model : Breakeven or zero-coupon inflation swap (bp) regressed against JPMCCI, JPMCCI-squared, 3Mx3M/15Mx3M OIS curve (bp), VIX Index, and Economists' inflation expectations (Survey of Professional Forecasters 10Yr US CPI - (0.908 + 0.277 2Yr US CPI + 0.38810Yr US CPI), %); model uses daily data over the past seven years

Source: J.P.Morgan

Seasonality in CPI and TIPS

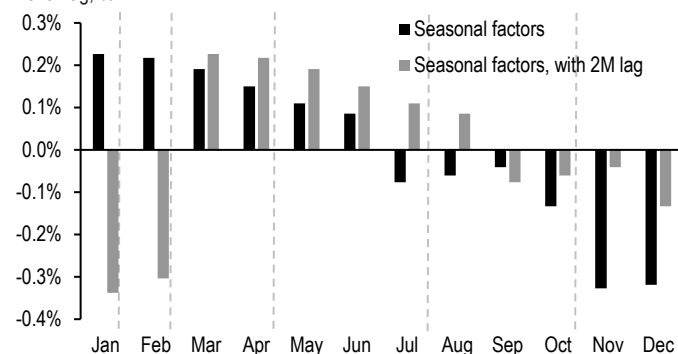
Headline CPI inflation tends to be strong in the first few months of the year, moderate in the middle of the year, and soften in the last quarter of the year. This comes from various seasonal factors that drive each of the CPI components—for example, energy price inflation tends to firm into the peak summer travel season before moderating, while retail discounts around the holiday season in November and December tend to push apparel prices lower, etc. The BLS uses statistical methods to derive these seasonal factors, and the seasonally-adjusted CPI is often a better indicator of underlying trends in inflation. However, TIPS are indexed to the unadjusted CPI, since seasonal adjustment factors are subject to revision for up to five years after their original

release. Thus, it is unsurprising that TIPS prices reflect the expected changes in CPI that normally occur at the same time each year.

Consider the relative pricing of securities maturing in different maturity months. **Figure 10** shows the additive 2025 seasonal adjustment factors from BLS by month—i.e., the difference between the m/m % changes in unadjusted and adjusted headline CPI. As mentioned, these seasonals tend to be very positive in the first few months of the year and very negative late in the year. Because of the lag in the indexation process, this means that July TIPS mature several months after positive seasonals, while January and April TIPS mature several months after negative seasonals. As a result, July TIPS trade rich relative to January and April TIPS with similar maturities. This effect is more pronounced for shorter-maturity TIPS: for example, a 0.50%-pt April-versus-July CPI differential is worth 26bp for Apr-28 and Jul-28 TIPS, but only 13bp for Apr-30 and Jul-30 TIPS (**Figure 11**).

Figure 10: Seasonality in headline CPI...

Difference between non-seasonally adjusted and seasonally-adjusted monthly changes in headline CPI based on 2025 BLS seasonal adjustment factors, and with 2-month lag; %



Note: Dashed lines denote when TIPS mature. These months are January, February, April, July and October. Issues maturing after positive seasonal factors, with 2M lag benefit from seasonals.
Source: Bureau of Labor Statistics, J.P.Morgan

Figure 11: ... causes July TIPS to trade rich to April and January TIPS

Estimated value of the seasonality difference* in return terms for TIPS of different maturity months and yield impact for TIPS impacts of different maturity years (units as indicated)

	Jan/Apr	Jan/Jul	Jan/Oct	Apr/Jul	Apr/Oct	Jul/Oct
Price impact (%)	0.12%	0.62%	0.64%	0.50%	0.52%	0.03%
Yield impact, by maturity year (bp of impact)						
2028	-7.4	-35.2	-34.6	-26.4	-26.5	-1.2
2029	-4.7	-22.5	-22.9	-17.4	-17.8	-0.8
2030	-3.4	-16.5	-16.8	-13.0	-13.4	-0.6

* Based on the values in Figure 10
Source: J.P.Morgan

Similarly, seasonality matters for the pricing of a given security at different times of the year. Consider a newly issued April-maturity 5-year TIPS: principal at maturity will be scaled based on CPI changes over five full years, so seasonality has no impact. As time passes from April through January, CPI (on a lagged basis) is seasonally positive. This suggests that if you purchase the security when it has 4.5 years to maturity, for example, seasonality is expected to pressure CPI lower over the final six months before maturity, so spot prices move lower (and yields higher) to reflect this. Conversely, from January to April, CPI is seasonally negative, so spot prices rise (yields fall). Since we can quantify these effects, we often find it useful to look at seasonally-adjusted TIPS yields and breakevens in order to assess performance and valuations and approximate market-based measures of expectations for trend inflation (see [How do you measure, measure a year?](#), 10/23/19).

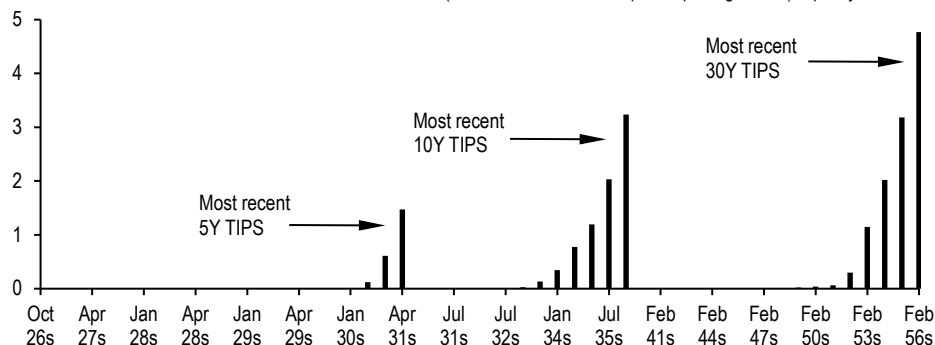
Value of the inflation floor in TIPS

When expected future inflation is well above zero, the embedded floor on TIPS principal (floored at par at maturity) is usually worth very little, because the option is

very far out-of-the-money. As **Figure 12** shows, the estimated values of the embedded floors currently range between 0-5bp of yield. These floors tend to decline in value as issues age and their index ratios rise above 1.0 (par) due to accrued inflation. The floors in recently-issued TIPS tend to be worth the most because their implied forward index ratios tend to be closest to 1.0. Additionally, TIPS with a greater time to maturity have a larger value given the time value of money for the embedded option.

Figure 12: The value of the embedded floor tends to be highest for recently-issued TIPS and increases with time to maturity

Estimated value of the embedded inflation floor in TIPS (based on our internal options pricing model); bp of yield

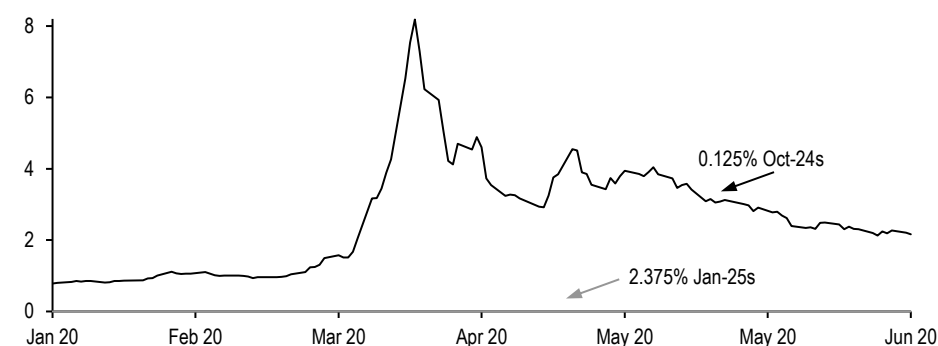


Source: J.P.Morgan

However, when expected future inflation is close to zero, the inflation floor on newly-issued TIPS can become quite valuable, as we saw during the financial crisis and COVID-19. In March 2020, inflation swap rates dropped sharply as COVID-19 spread through the US: **Figure 13** shows that the floor value on hot-run Oct-24s TIPS rose, causing Oct-24s TIPS to outperform versus near off-the-runs with high index factors, such as the 2.375% Jan-25s, though the move was not nearly as severe as we observed in late 2008.

Figure 13: The embedded floor premiums on low-index TIPS rose in March 2020

Premium of floor embedded in 0.125% Oct-24 TIPS versus floor in 2.375% Jan-25 TIPS; bp of yield



Source: J.P.Morgan

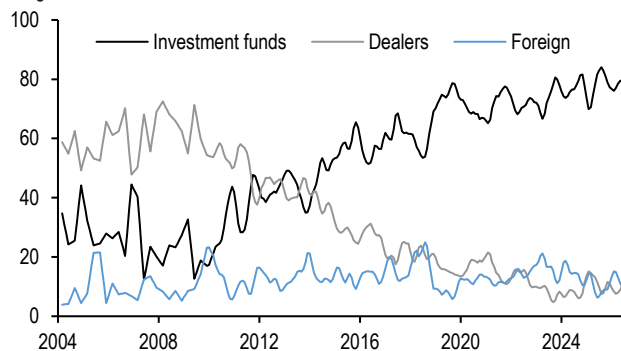
Who trades inflation?

In the early years of the TIPS program, dealers were the largest buyers of TIPS at auction, but over the last 15 years, end-user demand for TIPS has steadily increased (**Figure 14**). Specifically, investment-manager demand for TIPS has risen since the financial crisis. TIPS are *not* included in commonly-followed benchmark aggregate indices, but some core bond funds using these indices may still choose to make strategic

allocations to TIPS. In addition, some investment managers may have dedicated TIPS or inflation-protected funds and may buy (sell) TIPS when these funds see inflows (outflows) in order to match the profile of their benchmark index. As **Figure 15** shows, assets under management for inflation-protected bond funds grew sharply after the inflation shock, peaking post-COVID-19 in December 2021 at \$416bn before declining to \$300bn in October 2023 and then rising more recently after a wave of global tariffs and the conflict between the US and Iran in 2026. Inflows have been predominantly focused in short-dated funds in recent years, and our work has shown that most inflation-protected funds maintain an underweight TIPS exposure, often choosing to allocate out of benchmark in search of excess returns (see [The inflation fund landscape: reviewing trends in flows, composition, and positioning](#), 5/28/24). Our analysis also finds that flows into inflation-protected funds tend to follow the 1m lagged annual return of the TIPS asset class as measured by the J.P. Morgan TIPS Index (JUSTINE, **Figure 16**).

Figure 14: End-user demand for TIPS at auction has been increasing over the past two decades, driven by investment managers

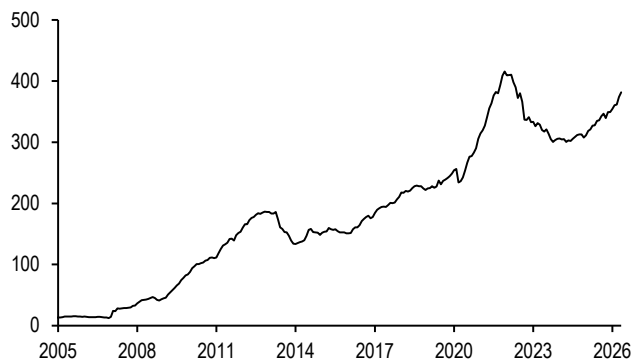
Allocation to dealers, investment funds, and foreign investors, 3-month moving average across all TIPS auctions; %



Source: US Treasury

Figure 15: Inflation-protected mutual funds grew sharply after a series of inflation shocks

Total net assets for inflation-protected domestic bond funds; \$bn



Source: EPFR

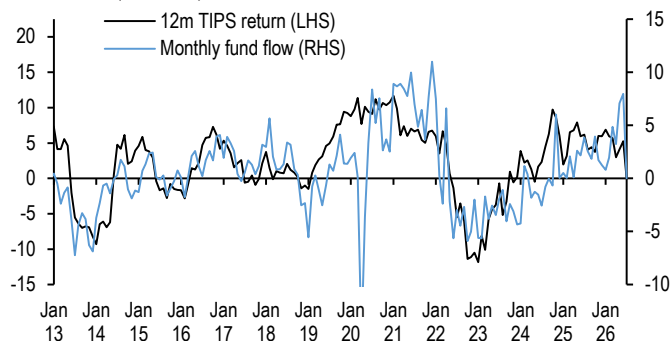
Although **foreign investors** buy a smaller portion of TIPS at auction, anecdotal evidence suggests that central banks and sovereign wealth funds are major players in the market. Traditionally, foreign central banks are mostly buy-and-hold investors, and operate with longer investment horizons than other real money investors. They may or may not manage their TIPS holdings against a benchmark index. Foreign holdings of TIPS are available annually from the Treasury International Capital System (TIC), and these data show that foreign investors accounted for 39% of the market as of June 2025, with China representing the largest share (**Figure 17**). TIPS now comprise roughly 11% of foreign official Treasury portfolios, the highest share on record. This reflects a broader trend in recent years whereby foreign official institutions have increased the proportion of their Treasury holdings dedicated to TIPS, FRNs, and short-term debt at the expense of long-term nominal holdings. Meanwhile, the foreign private and official institutional shares of the TIPS market have converged to roughly 19%, which represents a record high for the private community but represents a material decline for the official community from a peak of 32% in 2015.

Hedge funds are also large players in the TIPS market. They may use TIPS to express a view on real yields, the real curve, bond spreads, TIPS breakevens, and the breakeven curve. They may also opportunistically buy/sell TIPS to earn carry. Hedge funds are also very active in the very front end of the curve (Y to maturity). Real money accounts benchmarked to TIPS indices tend to sell their front-end TIPS as soon as they roll out of

the index, and hedge funds tend to buy this paper when it cheapens.

Figure 16: TIPS fund demand is driven by the asset class performance over the prior 12 months

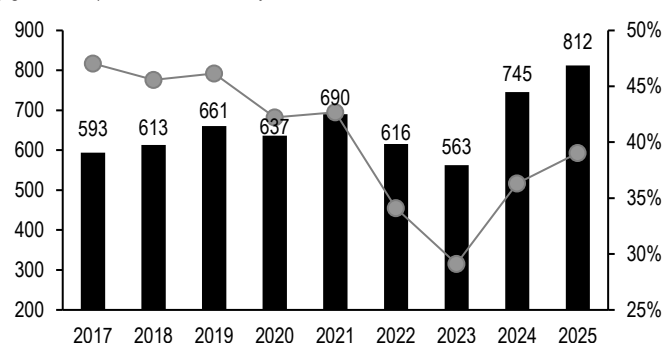
12-month TIPS return on our JUSTINE index (%; LHS) versus monthly inflows into US inflation funds (\$bn; RHS)



Source: EPFR, J.P. Morgan

Figure 17: Foreign holdings of TIPS account for 39% of the market and have been increasing over the past two years

Foreign holdings of TIPS (left axis; \$bn) and as a percentage of TIPS outstanding (right axis; %) as of June of each year

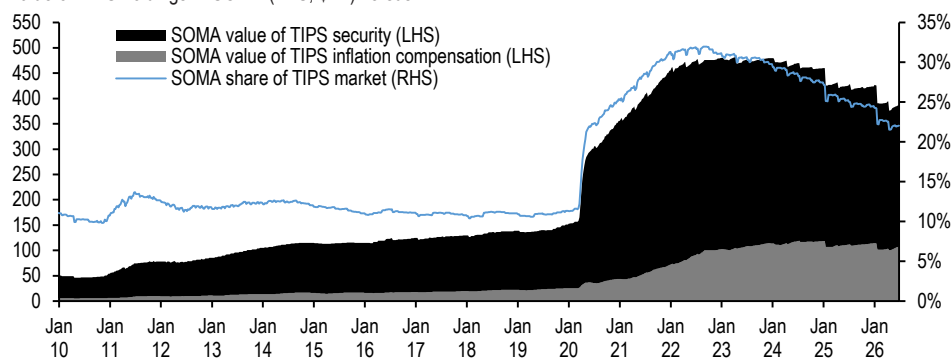


Source: Treasury International Capital System, US Treasury, J.P. Morgan

The **Fed** can be a large driver of demand for TIPS. Given the aggressive pace of Fed purchases throughout QE4, over \$482bn of TIPS (inflation-adjusted) were held in the SOMA portfolio at its peak in April 2023, representing more than 30% of TIPS outstanding (**Figure 18**). The peak in the SOMA share of the TIPS market was earlier in August 2022, just after the start of quantitative tightening in June 2022. Outside of another period of quantitative easing, we expect the Fed's share of the TIPS market to continue to decline as the Fed shifts the duration of its portfolio lower by increasing its bills share.

Figure 18: Given the aggressive pace of Fed purchases throughout QE4, over \$482bn TIPS (inflation-adjusted) were held in the SOMA portfolio at its peak in April 2023, representing more than 30% of TIPS outstanding

Value of TIPS holdings in SOMA (LHS; \$Bn) versus



Source: US Treasury, Federal Reserve Bank of New York, Haver Analytics

US CPI inflation swaps

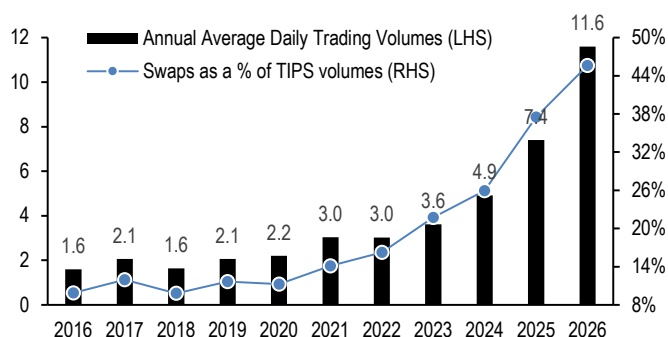
In Europe, inflation derivatives are more liquid and more widely-traded than cash linkers, but in the US, the opposite is true: the TIPS market is more liquid than US CPI derivatives. That said, the picture is shifting. Inflation swap trading volumes have expanded rapidly in the post-pandemic era of elevated inflation and heightened inflation uncertainty as investors have sought alternative ways to hedge inflation exposure and to trade their inflation views through more bespoke and flexible contracts. Indeed, **Figure**

19 shows that inflation swap daily trading volumes have averaged \$11.6bn in 1H26, up from \$2.2bn in 2020. Moreover, the growth in the inflation swap market continues to outpace that of the traditional cash market, with inflation swap activity now representing 46% of TIPS daily trading volumes, up from 11% in 2020. Inflation swap trading volumes remain concentrated in the intermediate sector of the curve, though growth in recent years has come predominantly from the front end (**Figure 20**).

This growth is also evident when looking across domiciles. As of June 2026, [LCH reported](#) \$16.6tn notional of inflation swaps outstanding (in USD-equivalent), up from \$7tn in the summer of 2020, and much of this growth has come from US inflation swaps. In terms of composition, USD swaps represent 34% of that total, up from a 24% share in 2020, while EUR and GBP swaps represent roughly 36% and 30%, respectively, down from each representing 38% in the summer of 2020.

Figure 19: Trading volumes in the US inflation swap market continue to grow rapidly on an outright and relative basis...

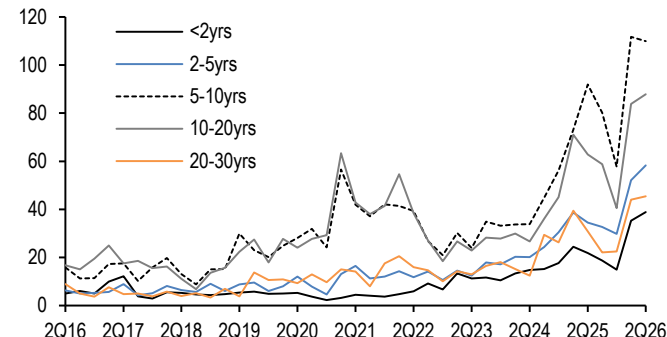
Annual average daily trading volumes of US CPI inflation swaps (LHS; \$bn) versus inflation swaps as a percentage of TIPS volumes* (RHS; %)



*2026 as of 6/30/26
Source: DTCC, J.P. Morgan

Figure 20: ...and trading volumes remain concentrated in the intermediate sector

Quarterly inflation swap trading volumes by maturity bucket; in \$bn 10-year CPI swaps equivalents*



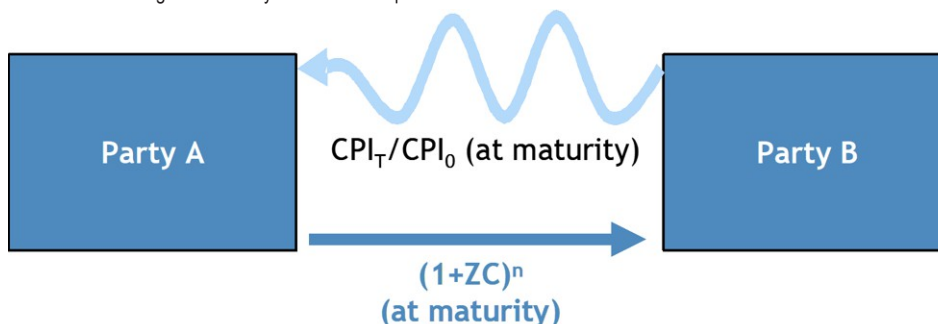
*Approximate, using PVBP at maturity bucket midpoints
Source: DTCC, J.P. Morgan

Zero coupon inflation swaps

Zero Coupon (ZC) inflation swaps are the most frequently traded US inflation derivative. The structure is very simple, with the only payments occurring at maturity (**Figure 21**). The party who receives inflation (pays fixed) in the swap will receive the accrued inflation over the life of the swap on the initial notional. The party that pays inflation (receives fixed) will receive a compounded fixed rate pre-agreed at the inception of the swap which represents the average annual inflation rate over the period. Two payments are made at the maturity of the swap, but in practice they are netted off. Importantly, the payment at maturity is not path dependent, depending only on the final reference CPI compared to the base CPI, both of which are subject to the same 3-month lag as TIPS. Thus, carry is largely irrelevant for zero-coupon swaps, though investors can still position for a slide down the curve. Also, due to the 3-month lag, it is important to look at 3-month forward-starting inflation swaps to examine benchmark inflation expectations (i.e. the 3mly inflation swap shows market expectations of inflation one year forward).

Figure 21: Zero coupon inflation swap cash flows

Cash flows exchanged at maturity of inflation swap



Source: J.P. Morgan

While ZC inflation swaps and TIPS securities reference the same BLS CPI NSA index, there have been episodes where the two have differed. In October 2025 a lengthy government shutdown precluded the BLS from publishing October CPI, forcing the two products to reference divergent fallback indices. TIPS followed Treasury's fallback, which is based on the 12th-root of the last published oya rate, whereas cleared ZC inflation swaps referenced ISDA's second fallback waterfall, which inflates the CPI value 12 months ago by the last published oya rate (see [TIPS Strategy](#), 9/26/25, and [TIPS Strategy](#), 10/24/25).

For zero-coupon swaps, breakeven inflation is the inflation that must be realized for the swap initiated at mid to realize zero value. This can be shown so that:

$$BEI = [(CPI_T / CPI_0)^{1/n}] - 1$$

where: CPI_T = CPI level upon maturity

CPI_0 = CPI level upon inception of the swap

n = tenor of the swap in years

BEI = Breakeven Inflation

Example: Suppose the current 10 year CPI inflation swap rate is 2.5%. Party A believes that annual CPI inflation will average more than 2.5% over the next ten years and thus purchases \$100mn of the zero-coupon swap, i.e. receive inflation. Assume the reference CPI index increases from 335 to 442 over 10 years. The total accrued inflation is thus $(442/335 = 1.31)$, which is equivalent to an annual inflation rate of 2.719% $(1.31^{1/10} - 1)$, which is above the initial fixed payment and thus Party A will net a gain over Party B. Specifically, the payment received by Party A from Party B on \$100mn notional is equal to:

$$(442/335) * \$100mn = \$131.9mn$$

The counterparty (Party B) will receive a compounded fixed rate, which was set at the inception of the swap and represents the expected breakeven inflation rate. Therefore, if the fixed rate in this case was 2.5%, the fixed payment Party B will receive from Party A will be equal to:

$$\$100mn * (1 + 2.5\%)^{10} = \$128.0mn$$

Therefore, the net payment in this example is that the inflation receiver (Party A) will be paid \$3.93mn by the inflation payer (Party B) upon maturity.

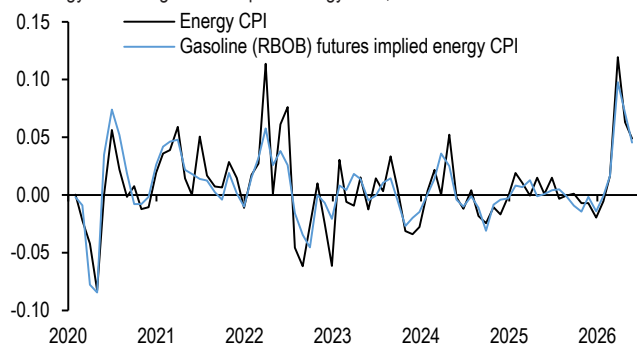
Trading CPI fixings

Investors can target an individual CPI fixing (i.e., the September 2026 CPI NSA level) rather than express a medium-term inflation view over the life of a TIPS or an inflation swap. This market is generally liquid only a few months into the future. If an investor “buys” the fixing, they make money if the realized CPI exceeds the quoted level, and vice versa if they sell. An investor buys the fix by entering a backward-starting one-year zero coupon swap that matures three months after the month in question. For example, to trade the September 2026 CPI fixing, an investor would receive inflation on a Dec’25/Dec’26 zero coupon inflation swap. If the risk in the trade is \$10,000 per 0.1-point deviation, the quoted level is 335.63, and the actual is 335.69, the payout is $(335.69 - 335.63) / 0.1 * (10,000) = \$6,000$.

Inflation swaps reference the CPI-U NSA index. However, market participants derive a market implied core fixing from the swaps market by subtracting energy and food inflation, given that core inflation can be more indicative of persistent trends in inflation. For the energy component, Gasoline makes up 47% of energy CPI, and we find that Gasoline RBOB explains most of the variance in monthly changes in energy CPI (**Figure 22**). We use the gasoline (RBOB) futures curve to project the forward price path of gasoline, then multiply the implied monthly changes in that path by energy’s weight in the CPI-U index to estimate its contribution to each CPI fixing. Subtracting this monthly contribution from the fixings curve creates a market-based measure of CPI-U ex energy (**Figure 23**). While it is tempting to model a true core CPI fixing by also subtracting out the impact of food inflation, we have not found a consistent way to derive a market implied food CPI from the futures market without incorporating our own subjective economic analysis.

Figure 22: Energy CPI has a strong relationship with gasoline futures

US energy CPI and gasoline-implied energy CPI*, %

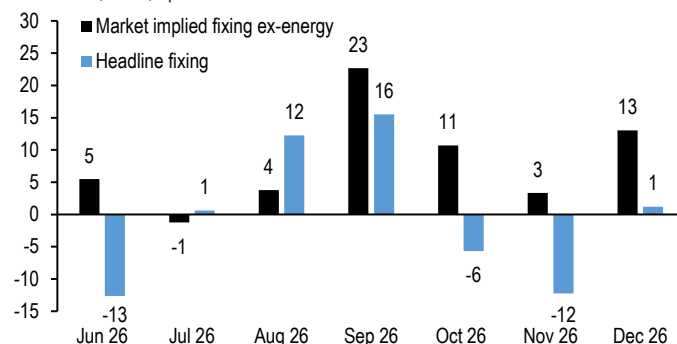


*Model $Y = 0.176X_1 + 0.095X_2 + 0.042X_3$; where X_1 = MoM change in daily avg. RBOB price and X_2 is and 1m lag of X_1 and X_3 is a 2m lag. $R^2 = 71\%$; $SE = 0.018$

Source: J.P.Morgan; Thomson Reuters; BLS

Figure 23: Market CPI fixings and derived fixings ex-energy

CPI-U NSA, MoM, bp



*Market Implied fixing ex-energy derived using model for energy CPI. Energy CPI nsa = $0.176X_1 + 0.095X_2 + 0.042X_3$; where X_1 = MoM change in daily avg. RBOB price and X_2 is and 1m lag of X_1 and X_3 is a 2m lag. $R^2 = 71\%$; $SE = 0.018$
Source: J.P.Morgan; Bloomberg Finance L.P.

Who trades inflation swaps?

Real money accounts generally receive inflation via swaps, either as an alternative to buying TIPS in an inflation-linked fund, or as an alpha-generation strategy in a core bond fund. They may also add a receive-inflation swap overlay to a muni or corporate bond position in order to create a synthetic inflation-protected asset. Occasionally, real money accounts may pay inflation in a swap as part of an alpha-generation strategy. Similarly, insurance companies generally receive inflation in order to hedge inflation-linked liabilities such as annuities. Occasionally they may buy TIPS on asset swap (which implicitly involves paying inflation) as an alternative to nominal Treasuries they otherwise need to buy. Hedge funds will trade both sides of the swap market (receiving

or paying inflation) and will often try to trade the range.

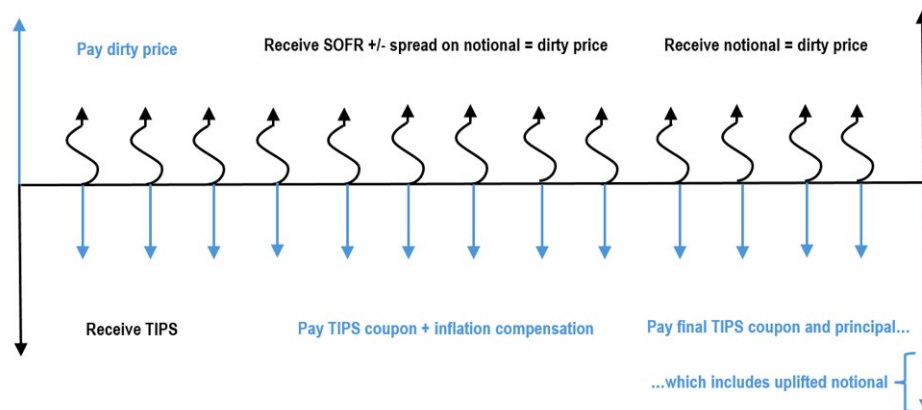
TIPS asset swaps

A traditional asset swap is a package that transforms a fixed-rate coupon bond into a floater that pays SOFR +/- a spread, plus exchanges of initial and final cash flows. The nature of the initial and final payment (par or dirty price) depends on the nature of the asset swap; i.e., if it's a par/par asset swap (par) or a proceeds-based asset swap (dirty price). In the Euro area, linkers typically trade on a par/par asset swap basis, but in the US, TIPS asset swaps are overwhelmingly proceeds asset swaps, and this is assumed to be the convention.

In a TIPS asset proceeds swap, the buyer first buys the bond at the current dirty price. Over the life of the TIPS, the investor pays all the TIPS cash flows to the dealer and receives SOFR +/- a spread on a notional equal to the initial dirty price. At maturity, the investor pays the dealer the final TIPS principal and final coupon and receives a notional equivalent to the initial dirty price (**Figure 24**). Since inflation is usually positive, the final TIPS cash flow will most likely be greater than the initial dirty price (i.e. notional uplift), so the final cash flow exchange will likely net out to a payment to the dealer. As a result, investors may incur credit/funding charges on longer-dated trades with a dealer.

Figure 24: A TIPS asset swap creates a synthetic floater

Cash flows exchanged between TIPS proceeds asset swap buyer and seller over the life of the TIPS*



*Perspective from TIPS ASW buyer. Black indicates receiving inflows, blue indicates paying outflows

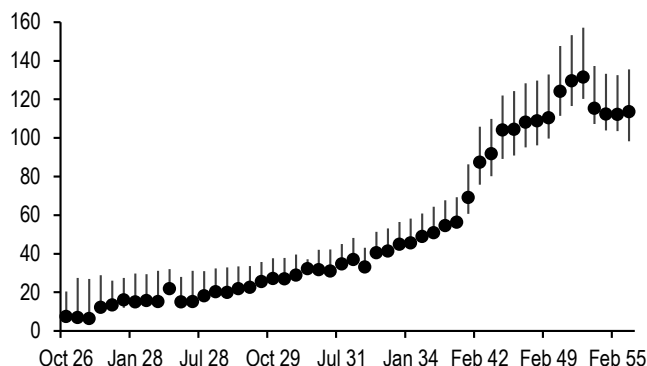
Source: J.P. Morgan

Note that asset swaps can be seen as a supply of inflation in derivatives/swaps form, as well as a supply of options/floors on the principal amount (given that in an asset swap the buyer gives the inflation-linked cash flows to the seller). On net, the asset swap removes duration, curve, and inflation risk, leaving a package which primarily represents credit and liquidity risk.

Hedge funds, insurance companies, and bank treasuries trade TIPS on asset swap. Some real money accounts may also participate. These investors are effectively getting paid for providing liquidity to the market. Investors may also buy TIPS asset swaps in order to realize roll-down, since the spread curve is generally upward sloping (**Figure 25**). They may also buy TIPS asset swaps in order to position for tighter spreads (i.e. given a view that TIPS should richen). Alternatively, investors may add a nominal interest rate swap to the TIPS asset swap in order to transform the package from a SOFR floater into a fixed-rate synthetic Treasury, with a pickup to nominal Treasuries. Doing so may allow some investors to use hedge accounting.

Figure 25: TIPS ASW allow investors to harvest the liquidity premium embedded in TIPS, take a view on spreads, and realize roll down

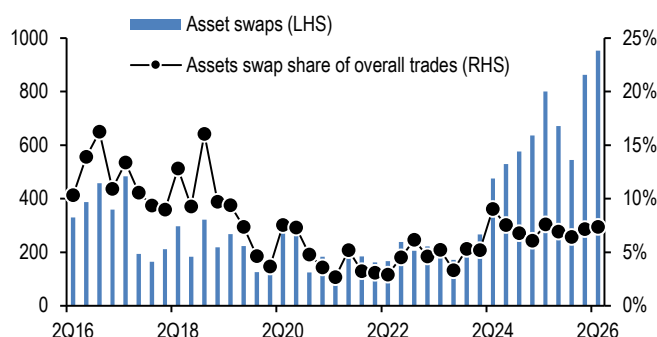
TIPS asset swap spread to SOFR across the curve within 6-month min/max range; bp



Source: J.P. Morgan

Figure 26: Asset swap transactions continue to represent roughly 7% of total inflation swap trading activity

Quarterly asset swap trade count (LHS; count) and asset swap share of overall trade count (RHS; %)*



* Assumes inflation swap transactions with a maturity date matching an outstanding TIPS reflect asset swap transactions
Source: DTCC, J.P. Morgan

The relative cheapness of TIPS asset swaps versus conventional Treasuries has made them the target of investors looking for attractive pickup. The (TIPS asset swap) minus (Treasury asset swap) differential is generally directional with the inflation basis. This means that when TIPS richen relative to Treasuries on asset swap, breakevens tend to widen relative to inflation swaps. This differential tends to be driven primarily by flows in the short run, and often sharp moves become exacerbated before the basis reverts to the mean.

In terms of trading volumes, we make the simplifying assumption that the vast majority of inflation swap transactions with a maturity date matching an outstanding TIPS reflect asset swap transactions. **Figure 26** shows that asset swaps have grown proportionally with the increase in overall inflation swap trading activity, and continue to represent 7% of total inflation swap transactions in 2Q26.

Inflation options: caps & floors

The market trades floors and caps on US CPI inflation. There is an array of ways to trade caps and floors through swaptions, but the primary two types of caps/floors that trade in the inflation market are Zero Coupon (ZC) and Year-over-Year (YoY). ZC caps/floors reference the cumulative inflation over a certain period, usually more than one year, while YoY caps/floors consist of a strip of independently-exercisable options which reference year-over-year inflation rates and have an annual, semiannual, or quarterly look. For example, consider a 10-year 0% YoY floor between 2026 and 2036. If realized inflation is +2% every year to 2036 with the exception of 2030-2031, when it is -5% (deflation), the payout is 5% for the owner of this floor. In the case of a 10-year 0% floor (ZC), however, the payout would be 0.

Zero coupon caps/floors trade in greater volume, especially floors with 0% strikes, since TIPS are a natural source of these floors. Year-over-year caps/floors trade less frequently, and premiums tend to be much higher because vol is much higher. The product was more popular in the early part of the last decade, with supply/demand coming from the structured note market and occasionally from corporate contracts linked to CPI, though trading of these types of notes slowed to a trickle over recent years. For both types of options, far out of the money strikes (0% and 4% for example) are more liquid than ATM strikes (1.5%-2.5%).

In general, the inflation option market is even less liquid than the inflation swap market, with virtually no inflation options traded in the interdealer market over the past three years, though volumes have started to pick up somewhat in recent months. Hedge funds and insurance companies are usually interested in buying options. Real money accounts are generally long TIPS in their portfolios, so they may sell inflation floors as a way to monetize the embedded optionality in TIPS and earn carry. They may also sell inflation caps as a covered call-like strategy.

Relative value

While the most common approach to compare TIPS is pricing on an asset swap basis, we also review here a number of other ways to investigate relative value among TIPS securities, including the inflation basis/IOTAs, using the par curve, and implied breakevens. We conclude with a brief discussion of the inflation swap curve.

The inflation basis and IOTAs

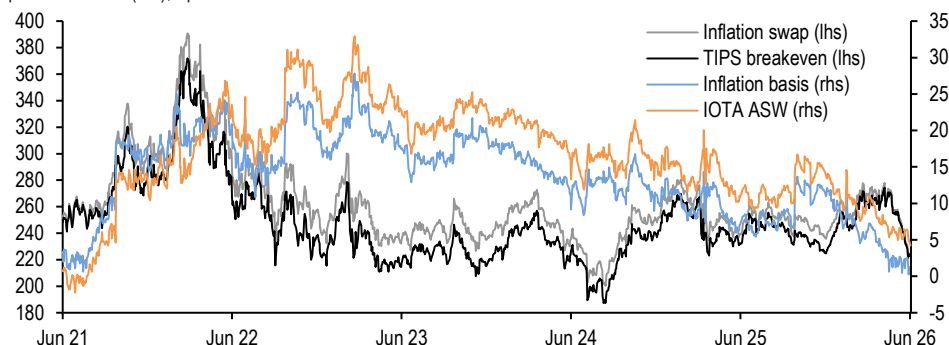
The difference between the inflation swap rate and the on-the-run TIPS breakeven is called the inflation basis. Positioning long the basis (i.e. wideners) involves going short breakevens (i.e. narrowers) and long inflation swaps (i.e. wideners). In theory, the inflation swap rate should be a cleaner representation of inflation expectations (as well as inflation risk premia), jettisoning the seasonality, carry, supply and demand imbalances, and balance sheet considerations that affect breakevens. As such, the inflation swap basis should reflect the relative liquidity premium embedded in TIPS, and can be a useful gauge of relative value for TIPS.

Indeed, as Figure 27 shows, inflation swaps and TIPS breakevens tend to be very well-correlated, but the inflation swap rate tends to be higher than the corresponding TIPS breakeven rate. One explanation for this consistently-positive basis is that the inflation swap market is predominantly one-sided: most investors want to receive inflation/pay fixed, and there are no natural payers of inflation (other than the Treasury through TIPS). As a result, the fixed rate that investors must pay is biased higher. One could argue that there is a credit component embedded in the swap rate as well: although the vast majority of inflation swaps are cleared, the counterparty is not the US government. In general, the basis tends to be mean-reverting over the medium term but very flow-driven over the short term. That said, the basis has been narrowing towards its tightest levels in recent years, which could be reflective of growing liquidity in the TIPS market, widening breakevens, and similarly increased liquidity and willingness from investors to pay inflation in inflation swaps, narrowing swaps.

Investors can trade the basis directly, but basis trades are less common than outright inflation swap trades. Historically, our basis models have tended to be unstable, with shorter-term correlations not tending to persist over longer horizons. That said, we would expect the basis to be at least partially explained by the cost and availability of balance sheet and overall TIPS market liquidity, and for the basis to trade directionally with nominal swap spreads. We also find that the basis tends to widen as the term GC/OIS spreads widen. Finally, we observe some seasonality in the basis, with cash breakevens tending to outperform swaps when ex-ante carry is higher.

Figure 27: Inflation swaps often trade above TIPS breakevens, though the “basis” has been narrowing recently

5-year US CPI inflation swap rate (lhs), 5-year TIPS breakeven (lhs), inflation swap basis (rhs) and IOTAs derived from proceeds ASW (rhs); bp



Source: J.P. Morgan

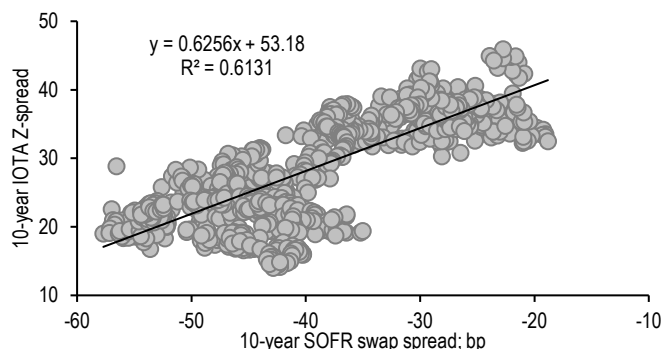
It's also important to note that the shape of the curve will influence the basis, since the inflation swap has a constant maturity, while the maturity of the hot-run TIPS will be shorter. Moreover, since the ZC inflation swap does not have an embedded floor, a richening or cheapening in the floor can also impact the basis. For these reasons, when gauging the relative performance of swaps versus breakevens, it is often useful to examine IOTAs, which compare breakevens with *matched-maturity* swap rates and adjust for the value of the floor on the TIPS. The higher the IOTA breakeven, the richer inflation swaps trade relative to cash breakevens.

We can observe IOTAs cleanly through asset swaps, where an investor buys TIPS on ASW and sells Treasuries on ASW, on a matched maturity basis. In the combined package, the investor's SOFR swap exposure is canceled, leaving the investor paying TIPS real yields and receiving Treasury nominal yields, which is equivalent to positioning short breakevens, while receiving inflation via the TIPS ASW. Thus, in effect, the investor has entered into a long IOTA position, looking for a weakening in the liquidity of TIPS versus Treasuries while hedging duration, curve, inflation, and credit components of the trade. As a way to harvest carry, IOTA spreads can trade directionally with swap spreads and are impacted by changes in volatility and funding conditions (**Figure 28**). The higher the IOTA ASW, the cheaper the TIPS is relative to the Treasury comparator.

Another way to inspect IOTAs is from Z-spreads. We can utilize inflation expectations embedded in inflation swaps to derive future index ratios, which we then apply to future coupon and principal payments of a TIPS to create a stream of inflation index-adjusted cash flows. Setting these nominal cash flows equal to the nominal dirty price of a TIPS, we can compute the synthetic nominal yield to maturity of a TIPS. The Z-spread between these synthetic nominal TIPS yields and our Treasury par curve thus theoretically captures the liquidity discount priced into TIPS. By applying the inflation trajectory implied by the inflation swap curve to each cash flow on a TIPS bond, Z-spreads more precisely capture the differential between cash breakevens and inflation swaps, even if they are not traded (see [TIPS synthetic nominal yields and Treasury z-spreads](#), 4/21/20 for more details).

Figure 28: IOTAs are well correlated with UST asset swap spreads

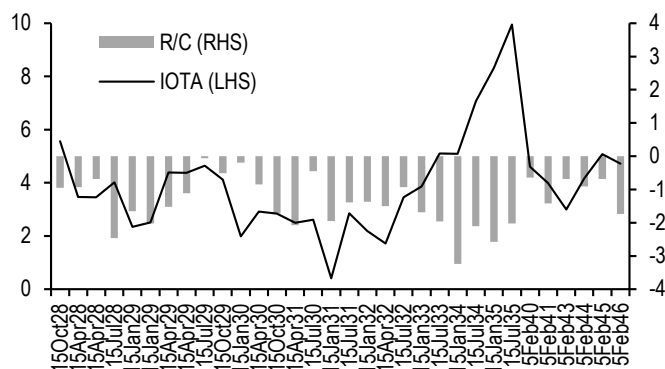
10-year IOTA derived from Z-spread regressed on UST SOFR swap spreads; regression over the past four years; bp



Source: J.P. Morgan

Figure 29: Using IOTAs, 2031 TIPS appear cheap along the curve, and recent outperformance has been concentrated in the 2034-35 bucket

IOTA Z-spread (lhs; bp) versus Rich/Cheap measure defined as current minus 3-month moving average (rhs; bp)



Source: J.P. Morgan

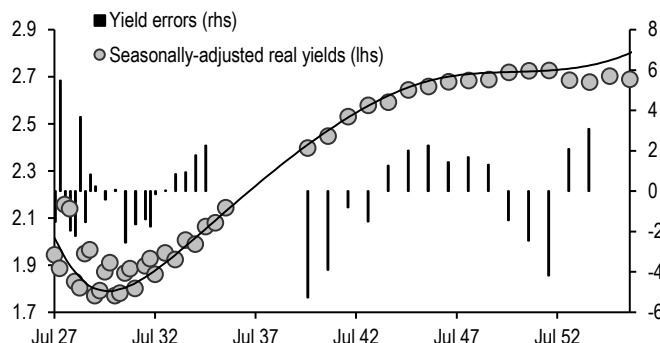
IOTAs are a useful metric to inspect security-specific TIPS relative value given their ability to strip away much of the duration, inflation seasonality, and credit concerns imbedded in real yields. Looking at IOTAs across the curve, **Figure 29** illustrates that TIPS in the 2031 bucket have richened significantly relative to the surrounding maturities. Indeed, defining the R/C measure as the difference between the current IOTA and its 3-month moving average, we can see that breakevens have been richening across the curve relative to inflation swaps, but not in a uniform manner, which may present relative value opportunities. Similarly, IOTAs can be a powerful tool to analyze the pricing of specific securities around certain key events (such as supply dynamics around auctions and index dynamics around month-, quarter-, and year-end).

Par curve and implied breakevens

Along a similar vein, we can inspect security-specific TIPS relative value through the fit to a par curve. We use a similar approach to the piece-wise cubic spline model utilized for nominal Treasuries, but we first adjust for the value of the TIPS floor value through a standard swaption pricing model and adjust for the seasonality of CPI using multiplicative seasonal factors from the BLS. We also filter out TIPS with less than 1-year to maturity from the estimation as they can distort curve fitting (see [The \(par\) curves they are a-changin'](#), 7/23/24 for more details). The dispersion of real yields relative to the par curve, which we define as yield errors, thus becomes a tool for screening relative value (**Figure 30**). Furthermore, the aggregate root mean squared error (RMSE) across all TIPS provides a helpful indicator of liquidity in the market, with lower values associated with a tighter fit to the par curve and thus increased TIPS liquidity.

Figure 30: The TIPS par curve can also be a useful measure of relative value across the curve

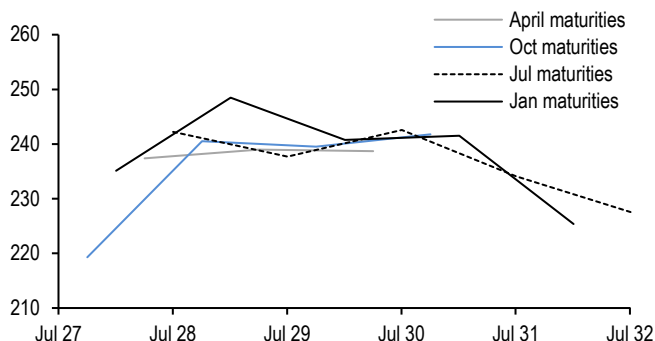
TIPS real yield par curve and seasonally-adjusted yield errors by maturity (lhs; %) versus yield errors (rhs; bp)



Source: J.P. Morgan

Figure 31: The Jan-29s appear slightly rich with respect to neighboring maturities

Forward 1-year breakevens bootstrapped from TIPS maturing in April, October, July, and January; bp



Source: J.P. Morgan

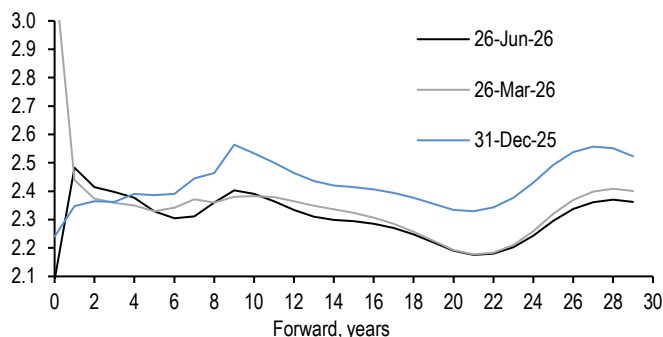
Turning to breakevens, another method to compare relative value across securities and strip out seasonality is by computing 1-year bootstrapped breakevens with the same maturity. Given the regular cadence of TIPS issuance, we can back out the implied 1-year inflation rate for each given security relative to the prior year's issue in order to avoid seasonality constraints. This analysis is most helpful at the front-end of the curve, where the aging of bonds provides a rich environment in which we can observe overlapping expectations between monthly vintages. **Figure 31** illustrates that the Jan-29s appear rich relative to path of inflation implied by surrounding maturities, potentially highlighting a relative value opportunity.

Inflation swap curve

Meanwhile, the flexibility of the inflation swap market makes it an ideal setting for looking at relative value across the curve. As inflation expectations are often autocorrelated (in the sense that forward inflation expectations are well correlated with the current rate of inflation) and they are often mean-reverting over time (if central banks keep to their inflation targets), dislocations in the inflation swap curve can be used for relative value trading. While the front-end of the curve is affected by near-term factors, such as fluctuations in energy prices, longer-dated forwards often retain a more stable shape over time (**Figure 32**). Thus, trading counter-directional forward-starting swaps (i.e. 2y3y/5y5y inflation swap curve steepeners when the curve appears historically flat), can be an important aspect of inflation market relative value (**Figure 33**).

Figure 32: The inflation swap curve has little slope to it, likely a combination of persistent inflation expectations and monetary anchoring

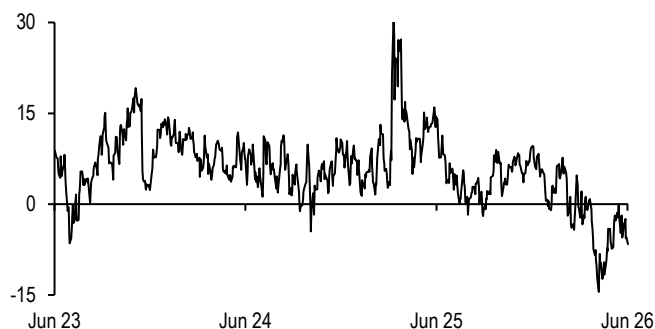
Forward 1-year CPI inflation swap rates; %



Source: J.P. Morgan

Figure 33: The 2y3y/5y5y inflation swap forward curve is inverted and trading near its flattest levels in recent history

2y3y/5y5y inflation swap forward curve; bp



Source: J.P. Morgan

Research and analytics

Readers can find many of these relative value measures within our *TIPS Relative Value Report* and *US Cash Interest Rate Analytics Package*. For J.P. Morgan's latest research and analytics reports on TIPS, please visit <https://markets.jpmorgan.com/jpmm/research.rates.inflation>

Appendix

Carry dynamics

One way of measuring carry is to take the forward real yield minus the spot real yield. If a linker is trading at a spot (clean) price P_s or spot real yield y_r we can calculate the carry of the linker using Equations 1-3 below. The carry in basis points would be $y_f - y_r$, the forward real yield minus the spot real yield.

Notation used in the formulae:

P_s = Real (clean) price of linker (quoted on market)

ct = Real coupon of linker at discrete times

y_r = spot real yield of linker (quoted on market)

I_s = Index ratio on spot date

I_f = Index ratio on forward date

is = Real coupon accrued as of spot date

if = Real coupon accrued as of forward date

$repof$ = repo rate for a term ending at the forward date, annualized

$\# \text{ days}$ = $\#$ days between spot and forward date

P_f = Real (clean) forward price of linker

y_f = Forward real yield of linker

Equation 1

$$P_s = \sum_{t=s} \frac{c_t}{(1 + y_r)^t} - i_s$$

Equation 2

$$P_f = \frac{I_s}{I_f} * (P_s + i_s) * (1 + repo_f * \frac{\#days}{360}) - i_f$$

Equation 3

$$P_f = \sum_{t=f} \frac{c_t}{(1 + y_f)^t} - i_f$$

If we look at a real example:

On May 29, 2026 (June 1, 2026 settle), the TII 1.875% Jan-36 traded at a spot real clean price of 98.4457, or real yield of 2.0534%. The index ratio at spot settle is 1.01624 and the spot real coupon accrued is 0.7096. On June 30, 2026, the index ratio is 1.02488, reflecting a 0.85% increase in CPI-U in April, and the real coupon accrued is 0.8598. We use 1-month General Collateral repo rate of 3.65%.

The spot cash dirty price would be (real clean price + real accrued interest)*(spot index ratio) = (98.4457+0.7096)*1.02488 = 100.766

Applying the formula above for the forward clean price in Equation 2:

$$P_f = (1.01624/1.02488)*(98.4457+0.7096)*(1+3.65\%*30/360)-0.8598 = 97.7586$$

Solving for yf in Equation 3, yf = 2.1326%

So given the forward and spot yields, the carry is:

$$2.1326\% - 2.0534\% = 7.9\text{bp}$$

Carry in an inflation breakeven trade: An inflation breakeven trade combines a long (short) position in a linker versus a short (long) duration-adjusted position in a similar-maturity nominal bond. As a result, the carry of such a trade can be thought of as the difference between the carry of the two legs. For example, if an investor goes long breakevens (buys TIPS and sells nominals), then the breakeven carry will equal (carry on TIPS) minus (carry on nominals). In general, carry on the trade will be positive when the accrued inflation (annualized) is higher than the level of the inflation breakeven.

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